Foundations of Statistical Learning & Algorithms

Today's goal • 3/3 items

Module 1
Introduction to Linear Algebra and Vector Spaces

Module 2
Linear Transformation and Matrix Decomposition

Linear Independence
- Understanding Linear Independence in Vector Spaces — Reading • 18 min
- Linear Independence, Basis, and Rank — Ungraded App Item • 20 min
- Linear Independence — Reading • 30 min

Linear Mapping
- Exploring Transformations: Understanding Linear Mappings — Reading • 1h
- Linear Mapping: Part 1 — Video • 8 min
- Linear Mapping: Part 2 — Video • 8 min
- Linear Mapping — Reading • 5 min
- Matrix Representation — Reading • 6 min
- Practice Quiz: Linear Mapping — Graded Assignment • Grade: 100%

Lesson 3: Eigenvalues and Eigenvectors
- Introduction to Eigenvalues and Eigenvectors — Reading • 40 min
- Eigenvectors and Eigenvalues: Examples and Applications — Reading • 30 min

Lesson 4: Cholesky Decomposition
- Cholesky Decomposition — Reading • 5 min
- Solved Examples — Reading • 10 min

Lesson 5: Singular Value Decomposition
- Introduction to Singular Value Decomposition (SVD) — Reading • 28 min
- Derivation of Singular Value Decomposition (SVD) from Eigenvalues and Eigenvectors — Reading • 15 min

Module 3
Probability Foundations for Statistical Learning

Lesson 1: Sum Rule, Product Rule, and Bayes' Theorem
- Sum Rule, Product Rule, and Bayes' Theorem — Reading • 15 min
- Sum Rule — Reading • 4 min
- Product Rule (Chain Rule) — Reading • 2 min
- **Bayes' Theorem** — Reading • 43 min

Lesson 2: Gaussian Distribution
- Univariate Gaussian Distribution — Reading • 4 min

Bayes' Theorem

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Bayes' theorem is a powerful tool in probability theory that allows us to update the probability of a hypothesis given new evidence. It relates the conditional probability of an event given a hypothesis to the conditional probability of the hypothesis given the event.

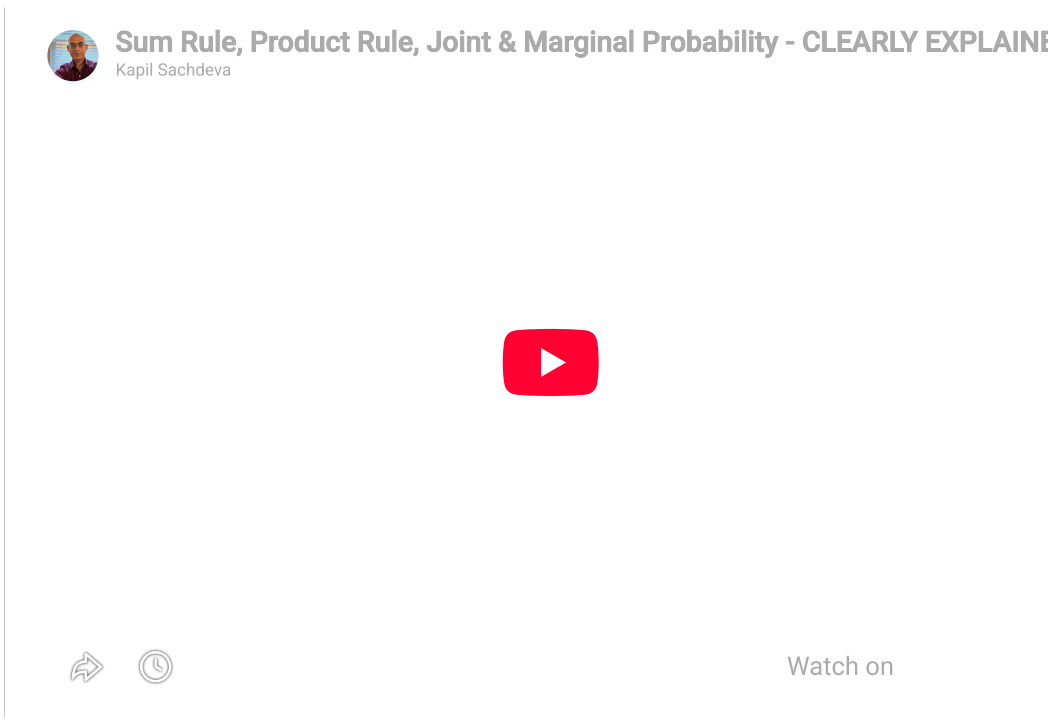
Formula

$$P(H \mid E) = \frac{P(E \mid H)P(H)}{P(E)}$$

where:

- H is the hypothesis
- E is the evidence or observed data
- $P(H \mid E)$ is the posterior probability of the hypothesis given the evidence
- $P(E \mid H)$ is the likelihood of the evidence given the hypothesis
- $P(H)$ is the prior probability of the evidence
- $P(E)$ is the marginal likelihood or evidence, often computed using the law of total probability

Through this video, you'll gain an intuitive and practical understanding of essential probability rules, like joint, marginal, and conditional probabilities. We'll work through examples with two buckets of fruit to build on these concepts, introducing Bayes' rule and how it's applied in probability-driven modeling.



In this video, you'll explore Bayes' theorem through its simplest case, uncovering how it helps us update probabilities based on new information.

